Latent Ordinal Evidence, Misaligned Outputs: Inference-Time Ordinal Lens Alignment for Multimodal LLMs

Haiming Li[*,1,2] **Yingsheng Liu**[*,1,2,†,‡] **Jingmin Zhu**[*,1,2] **Siyuan Yan**[1,2]
Xieji Li[1,2] **Jiajun Sun**[2] **Zongyuan Ge**[1,2] **Zhen Yu**[1,2,‡]

[1]Faculty of Information Technology, Monash University, Melbourne, Australia
[2]Monash University, Melbourne, Victoria, Australia

Correspondence: Yingsheng.Liu@monash.edu, Zhen.Yu1@monash.edu

Abstract

Multimodal LLMs apply the language model interface to visual inputs, where ordinal regression tasks such as age estimation, image quality assessment, and disease grading require autoregressive decisions over ordered class labels. We ask whether MLLMs reliably convert internal ordinal evidence into ordered digit-token outputs. Across four ordinal benchmarks and four MLLM backbones, ordinal labels are linearly recoverable from hidden states with Spearman correlation up to 0.938, and a task-designed prompt further sharpens this structure. Yet native digit-token outputs weakly expose it: the unembedding matrix filters the ordinal direction, and the digit-token row space retains below 1.15% across all 16 model-dataset combinations, with a 16 to 77 absolute-point accuracy gap between linear-probe and native outputs. We introduce Ordinal Lens Alignment (OLA), a frozen-backbone inference-time method that trains lightweight W_S-anchored lenses on mid-to-deep decoder layers, fuses them into an ordinal distribution, and corrects only digit-token logits at generation. OLA outperforms the SOTA LoRA-tuned OrderChain baseline in most settings while keeping the MLLM frozen, surpasses discriminative ordinal baselines in most cells, and improves over an offline lens in every setting.

1 Introduction

Ordinal prediction differs from nominal classification because its errors carry distance. Misclassifying age group 3 as group 4 differs qualitatively from misclassifying it as group 7, and similar gradations arise in disease grading, image quality assessment, and aesthetic rating. Discriminative ordinal regression has been addressed through rank-aware decoders (Cheng et al., 2008; Niu et al., 2016; Cao et al., 2020; Shi et al., 2023), label-smoothing and diversity objectives (Diaz and Marathe, 2019; Berg et al., 2021), domain-specific aesthetic and medical scoring (Talebi and Milanfar, 2018; Jia et al., 2017), and vision-language formulations (Du et al., 2024; Li et al., 2022; Wang et al., 2023). An alternative route reformulates the task as digit-token generation through prompted MLLMs (Wang et al., 2025); this is attractive because it preserves the language-model generation interface and supports rationale-style responses. However, autoregressive token decoding over a discrete vocabulary provides no architectural guarantee that the generated digit tokens respect the ordinal scale of the underlying labels: a language model can emit a valid token without emitting an ordinally appropriate one, a tendency recently audited as central-tendency bias in MLLM clinical ordinal scoring (Zhang et al., 2026).

We examine this issue through a diagnostic study across four ordinal benchmarks (Adience, Diabetic Retinopathy, Historical Color Image, and Aesthetic) and four MLLM backbones (Qwen2.5-VL, Qwen3-VL, Gemma-4, and LLaVA-NeXT). Three observations emerge. First, hidden states contain strong ordinal evidence: linear probes recover ordered labels with peak Spearman correlation up to 0.938 (Table 1), and a task-designed prompt further sharpens this structure in last-token representations. Second, native digit-token outputs do not reliably expose this evidence: the unembedding matrix W_U spectrally filters the probed ordinal direction, and the digit-token row space W_S retains less than 1.15% of its energy across all 16 model-dataset combinations (Figure 1, Table 2). Third, this output-side weakness corresponds to a behavioral accuracy gap of 16 to 77 absolute points between linear probes on hidden states and the native digit-token output, indicating that what is linearly decodable internally can differ systematically from what the generation interface produces.

Code: https://github.com/Ethan-ysliu/OLA
[*] **Equal Contribution.**
[†] **Project Leader.**
[‡] **Corresponding Author.**

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing, pages 21328–21346
October 24-29, 2026 ©2026 Association for Computational Linguistics

Backbone	Adience ($C=8$)			DR ($C=5$)			HCI ($C=5$)			Aesthetic ($C=5$)		
	IT/N	LT/N	LT/D	IT/N	LT/N	LT/D	IT/N	LT/N	LT/D	IT/N	LT/N	LT/D
Qwen3-VL	0.916	0.872	**0.938**	0.427	0.276	**0.758**	0.617	0.556	**0.843**	0.859	0.864	**0.916**
Gemma-4	0.887	0.893	**0.929**	0.362	0.255	**0.757**	0.430	0.551	**0.773**	0.794	0.838	**0.919**
LLaVA-NeXT	0.917	0.920	**0.924**	0.421	0.365	**0.756**	0.554	0.586	**0.800**	0.822	0.860	**0.906**

Table 1: SRCC Diagnostics for Probe-Readable Ordinal Evidence

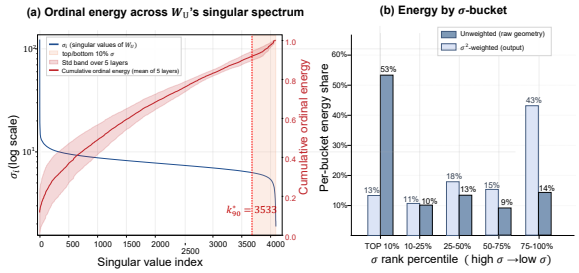


Figure 1: Spectral Filtering of the Probed Ordinal Direction by the Unembedding Matrix W_U

Backbone	Adience	DR	HCI	Aesthetic
Qwen2.5-VL	1.05%	0.92%	0.99%	1.01%
Qwen3-VL	1.12%	1.06%	1.08%	1.14%
Gemma-4	0.97%	0.88%	0.93%	0.96%
LLaVA-NeXT	1.03%	0.95%	0.97%	1.02%

Table 2: Digit-Token Row-Space Retention of Probe-Derived Ordinal Directions

Motivated by this gap, we propose Ordinal Lens Alignment (OLA), a frozen-backbone inference-time method that bridges latent evidence to digit-token logits without modifying the vision encoder, language-model backbone, or unembedding matrix. OLA trains lightweight W_S-anchored lenses on K mid-to-deep decoder layers, fuses them via a learned softmax weighting into an instance-specific ordinal distribution P^{OLA}, and applies a target-restricted, discrepancy-driven correction to digit-token logits at the answer position.

Our contributions are threefold. First, we provide a diagnostic account that separates probe-readable latent ordinal evidence from its weak exposure in native digit-token outputs across four backbones and four benchmarks, with SRCC up to 0.938 yet below 1.15% W_S row-space retention in all 16 combinations. Second, we propose Ordinal Lens Alignment (OLA), a frozen-backbone inference-time method that connects multi-layer W_S-anchored lenses to digit-token logits through a target-restricted, discrepancy-driven correction, without modifying the vision encoder, language-model backbone, or unembedding matrix. Third, OLA improves over prompt-only, hidden-steering, capacity-matched, and offline-lens alternatives, and outperforms the SOTA LoRA-tuned OrderChain baseline in most settings while requiring only lightweight lens training without LoRA fine-tuning.

2 Task Setup & Diagnostic Findings

2.1 Task setup

We consider visual ordinal prediction with MLLMs. An input (x, P) pairs an image with a prompt specifying the task and answer format; the label $y \in \{0, 1, \ldots, C-1\}$ is ordered, with C varying across benchmarks. The MLLM predicts via greedy autoregressive decoding; $\hat{y} = c$ iff the emitted digit token equals s_c, where $\mathcal{S}_{\text{digit}} = \{s_0, \ldots, s_{C-1}\}$ is the per-tokenizer digit-token set. Let $h_\ell(x, P) \in \mathbb{R}^d$ be the last-token hidden state at layer ℓ, $W_U \in \mathbb{R}^{|\mathcal{V}| \times d}$ the unembedding matrix, and $W_S = W_U[\mathcal{S}_{\text{digit}}] \in \mathbb{R}^{C \times d}$ its digit-token row submatrix. Evaluation uses ACC, MAE, and SRCC: top-1 correctness, ordinal error magnitude, and monotonic agreement of predicted or probed indices with ground-truth labels.

We evaluate on Adience (Eidinger et al., 2014) ($C = 8$), DR (Dugas et al., 2015), HCI (Palermo et al., 2012), and Aesthetic (Schifanella et al., 2015) ($C = 5$ each) with the open-source MLLM backbones Qwen2.5-VL (Bai et al., 2025b), Qwen3-VL (Bai et al., 2025a), Gemma-4 (Team et al., 2026), and LLaVA-NeXT (Liu et al., 2024); SRCC diagnostics use the latter three as representatives. Per-layer linear probes on frozen hidden states at three diagnostic sites yield the maximum test SRCC across layers, most recoverable in mid-to-deep layers, suggesting ordinal structure emerges after cross-modal fusion but is weakly exposed at the digit-token interface. Dataset descriptions, class distributions, and probe-site protocols appear in Appendix A.

2.2 Latent Ordinal Evidence Is Probe-Readable

For each diagnostic site and decoder layer, we train a free linear readout over frozen hidden states, in the spirit of linear probing (Alain and Bengio, 2016; Hewitt and Manning, 2019) and lens-based interpretation (Belrose et al., 2023; Park et al., 2023; Li et al., 2026):

$$p_\ell^{\mathrm{pr}}(c \mid x, P) = \sigma\Big(B_\ell \tilde{h}_\ell(x, P) + a_\ell\Big)_c \quad (1)$$

where $\sigma(\cdot)$ denotes the softmax and $\tilde{h}_\ell = (h_\ell - \mu_\ell)/\sigma_\ell$ is standardized. The free probe is unconstrained by native digit-token rows, testing whether ordinal labels are linearly recoverable from frozen representations. Table 1 reports the peak test SRCC across decoder layers under the three protocols of Section 2.1; per-layer SRCC breakdowns across all three backbones are in Appendix D.1.

Under the most informative protocol (last-token hidden state with a task-designed prompt), every model-dataset cell reaches SRCC ≥ 0.756, peaking at 0.938 on Adience with Qwen3-VL, 0.919 on Aesthetic with Gemma-4, and 0.843 on HCI with Qwen3-VL. Across all 12 backbone-dataset combinations of Table 1, ordered labels are linearly recoverable from frozen hidden states, so downstream ordinal errors cannot be explained by absent internal evidence.

The task-designed prompt (specifying the ordinal scale and answer format; see Appendix B for templates and design principles) consistently improves last-token SRCC, with largest gains where the neutral probe is weakest: on DR, SRCC rises from 0.255–0.365 to 0.756–0.758; on HCI, from 0.551–0.586 to 0.773–0.843. Adience and Aesthetic already show strong neutral or visual-token recoverability, so prompt gains are smaller. The prompt thus helps the LM aggregate task-relevant cross-modal evidence at the answer position; whether the native digit-token interface uses it effectively is the focus of the next diagnostic.

Finding 1. MLLMs contain strong ordinal evidence in hidden states.

2.3 The Output Interface Filters and Weakly Exposes Ordinal Evidence

Finding 1 establishes that ordinal labels are linearly recoverable; we now ask whether the same evidence reaches the digit-token logits. We examine W_U from two complementary angles, its spectral structure and the digit-token row space W_S, and relate both to the behavioral gap between hidden-state probes and native digit-token outputs.

Let $W_U = U\Sigma V^\top$ with singular values $\sigma_1 \geq \cdots \geq \sigma_d$ and right singular vectors V_i. For a probe-derived ordinal direction v^{ord}, let $\alpha_i = \langle v^{\mathrm{ord}}, V_i \rangle$. Partitioning singular indices into rank buckets $\mathcal{B}$ (top 10%, 10–25%, 25–50%, 50–75%, 75–100%), we report two normalized energy shares per bucket:

$$E^{\mathrm{raw}}(\mathcal{B}) = \frac{\sum_{i\in\mathcal{B}} \alpha_i^2}{\sum_{j=1}^{d} \alpha_j^2}, \qquad E^{\mathrm{out}}(\mathcal{B}) = \frac{\sum_{i\in\mathcal{B}} \alpha_i^2\sigma_i^2}{\sum_{j=1}^{d} \alpha_j^2\sigma_j^2}. \quad (2)$$

E^{raw} measures the raw distribution of the probe direction across the singular spectrum, and E^{out} the distribution after the per-direction σ^2 weighting that $v \mapsto W_U v$ imposes. Figure 1 shows the top 10% bucket carries only 13% of the raw probe-direction mass yet contributes 53% of the σ^2-weighted output energy, while the bottom 25% carries 43% of the raw mass but only 14% after σ^2 weighting. The output interface therefore reweights the ordinal direction through the singular spectrum of W_U, attenuating a broad low-σ component and amplifying a smaller high-σ one, echoing structured signal patterns previously observed in LLM unembedding spectra (Cancedda, 2024).

Since ordinal prediction is evaluated through digit tokens, we measure the retention of v^{ord} in the digit-token row space W_S:

$$R_{W_S}(v^{\mathrm{ord}}) = \frac{\|\Pi_{\mathrm{row}(W_S)} v^{\mathrm{ord}}\|_2^2}{\|v^{\mathrm{ord}}\|_2^2} \quad (3)$$

where $\Pi_{\mathrm{row}(W_S)}$ is the orthogonal projector onto $\mathrm{row}(W_S)$. Table 2 reports R_{W_S} across four backbones and four benchmarks: cell means span 0.88%–1.14%, mean 1.00%. Only about one percent of the probe-derived ordinal direction is represented in the digit-token row space. This does not imply destruction of the remaining energy or that OLA modifies W_S; the native interface simply exposes this direction weakly. Behaviorally, this filtering aligns with accuracy gaps of 16 to 77 absolute points between hidden-state linear readouts and native digit-token outputs, largest on DR and HCI where native generation is weakest.

Finding 2. Native digit-token outputs weakly expose the recovered ordinal evidence.

Findings 1 and 2 identify an interface-level failure mode: evidence is present in frozen hidden states but only weakly exposed in the native digit-token logits. This mismatch does not dissolve with scale, since on a 32B backbone of the same family the ordinal direction remains linearly recoverable, the digit-token row space exposes an even smaller share of it, and native digit-token accuracy improves only marginally (Appendix E). The intervention should therefore connect probe-readable evidence to digit-token logits at generation, without teaching the backbone or retraining W_U; Section 3 introduces Ordinal Lens Alignment (OLA) to implement this.

3 Ordinal Lens Alignment

Ordinal Lens Alignment (OLA) is a frozen-backbone inference-time procedure that connects probe-readable ordinal evidence in mid-to-deep decoder layers to the digit-token logits at the answer position, without modifying the vision encoder, language-model backbone, or unembedding head. The only learned parameters are lightweight lens and fusion weights, trained once per backbone-dataset pair.

3.1 Overview

Figure 2 summarizes OLA as an output-side intervention at the answer-token decoding step. Given (x, P) with task-designed prompt P, the frozen MLLM produces native logits at the answer position. Forward hooks capture last-token hidden states at K mid-to-deep decoder layers $\mathcal{L} = \{\ell_1, \ldots, \ell_K\}$; each is passed through a lens mapping it into digit-token scores via the frozen W_S (Section 3.2). A learned softmax then fuses per-layer scores into an instance-specific ordinal distribution $P^{\mathrm{OLA}}(c \mid x, P)$ over $\mathcal{S}_{\text{digit}}$ (Section 3.3). Before greedy decoding selects the answer token, OLA compares P^{OLA} with the native digit-token distribution $Q(c \mid x, P)$ and applies a target-restricted, discrepancy-driven correction only to logits in $\mathcal{S}_{\text{digit}}$ (Section 3.3); other logits remain unchanged.

This decomposition separates three functions entangled in native MLLM output: latent evidence readout, multi-layer aggregation under a fixed K-layer window, and inference-time alignment with the digit-token interface. OLA combines logit-lens style readouts (Belrose et al., 2023) with inference-time output-side adjustment, unlike representation steering on hidden states (Panickssery et al., 2023) or LoRA-based MLLM adaptation (Wang et al., 2025); see Appendix H for extended discussion.

3.2 Ordinal Lens

The per-layer lens maps a frozen hidden state into a digit-token score vector. Following Findings 1 and 2, it is linear (ordinal labels are linearly recoverable) and anchored on W_S (native digit-token rows weakly expose the probe direction), expressing probe-readable evidence as digit-token scores rather than repairing W_S or the LM head.

Let $h_\ell(x, P) \in \mathbb{R}^d$ be the last-token hidden state at layer ℓ, captured during the same forward pass as the native logits. For each $\ell \in \mathcal{L}$, the lens computes

$$\hat{z}_\ell(x, P) = W_S\big(A_\ell \tilde{h}_\ell(x, P) + b_\ell\big) \tag{4}$$

with $\tilde{h}_\ell = (h_\ell - \mu_\ell)/\sigma_\ell$ and $A_\ell = I_d + U_\ell V_\ell^\top$. Here $\hat{z}_\ell \in \mathbb{R}^C$ is the digit-token score vector, $\mu_\ell, \sigma_\ell \in \mathbb{R}^d$ are channel-wise statistics on the training split, $U_\ell, V_\ell \in \mathbb{R}^{d \times r}$ define a low-rank residual transformation ($r \ll d$), and $b_\ell \in \mathbb{R}^d$ is a learned offset; trainable parameters are $\{U_\ell, V_\ell, b_\ell\}$. $W_S \in \mathbb{R}^{C \times d}$ is frozen, reused from the MLLM unembedding head.

Each layer lens is trained independently with cross-entropy over ordinal labels,

$$\mathcal{L}_{\text{lens}}^{(\ell)} = -\mathbb{E}_{(x,P,y)\sim\mathcal{D}_{\text{train}}} \log \sigma\big(\hat{z}_\ell(x, P)\big)_y \tag{5}$$

on cached hidden states, with no gradient flowing through the MLLM backbone or W_S. Unlike the diagnostic probe with a free matrix B_ℓ, the lens factors the readout through W_S, making it an interface module rather than a replacement classifier. Implementation details appear in Appendix C.

3.3 Aggregation and Online Alignment

The per-layer lens (Section 3.2) yields K complementary, W_S-anchored readouts from different mid-to-deep layers; OLA aggregates them into an ordinal distribution that corrects the native digit-token logits at generation.

OLA fuses the K per-layer scores via a learned softmax over layers, with $\alpha \in \mathbb{R}^K$ the fusion parameter and $a = \mathrm{softmax}(\alpha) \in \Delta^{K-1}$ the non-negative weights:

$$\begin{aligned} z^{\mathrm{OLA}}(x, P) &= \sum_{\ell \in \mathcal{L}} a_\ell \, \hat{z}_\ell(x, P), \\ P^{\mathrm{OLA}}(c \mid x, P) &= \sigma\big(z^{\mathrm{OLA}}(x, P)\big)_c. \end{aligned} \tag{6}$$

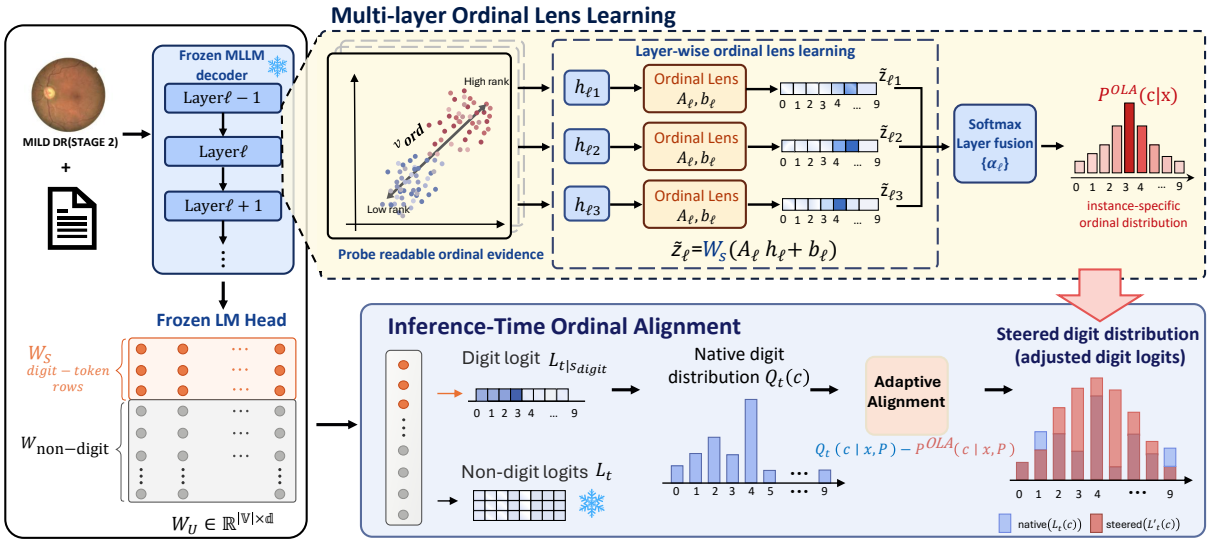


Figure 2: Architecture of Ordinal Lens Alignment for Inference-Time

α is learned on the training split with lenses frozen via the same cross-entropy objective; the aggregation adds only K scalars, independent of C and d.

At inference, the frozen MLLM produces native logit $L^{\text{na}}(c \mid x, P)$ for digit token s_c, $c \in \{0, \ldots, C-1\}$; we define the native digit-token distribution $Q(c \mid x, P) = \sigma(L^{\text{na}})_c$. Before greedy decoding emits the answer token, OLA applies a discrepancy-driven correction to the digit-token logits:

$$\begin{aligned} L'(c \mid x, P) = & L^{\text{na}}(c \mid x, P) \\ & - \lambda^\star \, \omega(P^{\text{OLA}}) \big(Q(c \mid x, P) \\ & - P^{\text{OLA}}(c \mid x, P)\big) \end{aligned} \quad (7)$$

for $c \in \{0, \ldots, C-1\}$, where $\lambda^\star \geq 0$ is selected on the validation split and $\omega(P^{\text{OLA}}) = \max_c P^{\text{OLA}}(c \mid x, P)$ is a confidence gate. The correction is sign-aware: it lowers $L'(c)$ when Q assigns more probability to c than P^{OLA} does and raises it otherwise. Logits outside $\mathcal{S}_{\text{digit}}$ (EOS and non-digit entries) remain unchanged; greedy decoding then emits the ordinal prediction. We correct digit-token logits rather than W_U because every ordinal decision passes through them, whereas editing W_U perturbs all tokens. The complete lens-training and inference-time alignment algorithms appear in Appendix G.

4 Experiments

This section evaluates OLA along the evidence chain of Section 2. The main benchmark in Section 4.1 tests whether OLA reduces the behavioral gap between probe-readable hidden-state evidence and native digit-token outputs identified in Section 2.3, comparing it with discriminative ordinal references and a LoRA-tuned MLLM baseline. The component ablation in Section 4.2 isolates which parts of OLA drive the improvement; Section 4.4 then examines ordinal output structure and hidden-representation geometry. Each result is tied to the diagnostic claim it tests.

4.1 Main Benchmark

We evaluate four MLLM backbones (Qwen2.5-VL, Qwen3-VL, Gemma-4, LLaVA-NeXT) on the four ordinal benchmarks of Section 2.1. Adience, DR, and HCI are reported as mean and standard deviation over five folds; Aesthetic uses the balanced test split. For all MLLM-based methods, image inputs, label mappings, digit sets, and evaluation parsers are held consistent within each backbone-dataset setting, while method-specific prompts and training procedures follow the baseline definitions. We compare OLA against three families. The first contains discriminative ordinal references (Ord2Seq, NumCLIP) trained outside the MLLM generation interface. The second contains frozen-backbone MLLM baselines: Naive Prompt for direct digit-token generation under minimal scaffolding, and CAA (Panickssery et al., 2023) for validation-selected residual-stream steering with a class-difference direction. The third is OrderChain (Wang et al., 2025), the published MLLM ordinal baseline applying LoRA fine-tuning (Hu et al., 2021) under a task-specific prompt convention. These comparisons span task-specific discriminative prediction,

frozen MLLM prompting, hidden-state steering, backbone adaptation, and the output-side alignment OLA introduces.

Table 3 reports the main benchmark (full mean±std over five folds in Appendix D.3). OLA improves over Naive Prompt in every model-dataset setting, with the largest gains on DR and HCI, where the diagnostics in Section 2.3 show a wide gap between probe-readable hidden-state evidence and weak native digit-token behavior. On DR, OLA raises Qwen2.5-VL from 0.156 to 0.920 ACC, Gemma-4 from 0.137 to 0.922, and LLaVA-NeXT from 0.148 to 0.924; on HCI, OLA raises Qwen3-VL from 0.328 to 0.757 and Gemma-4 from 0.245 to 0.733. These gains do not simply track zero-shot backbone strength; they are strongest where the native output interface underuses recoverable ordinal evidence.

Compared with CAA, OLA is higher in every model-dataset setting. CAA also intervenes on hidden representations but injects a single residual-stream direction and relies on the native LM head to convert that perturbation into a digit-token decision. OLA instead reads ordinal evidence through W_S-anchored lenses and applies a target-restricted correction directly to digit-token logits. This supports the diagnostic from Section 2.3: a strong hidden-state direction alone does not ensure reliable ordinal transfer through the native output interface.

Compared with the SOTA LoRA-tuned OrderChain baseline, OLA achieves higher ACC in 14 of 16 model-dataset settings while keeping the vision encoder, language-model backbone, and unembedding matrix frozen. The two exceptions are Qwen2.5-VL and Qwen3-VL on Aesthetic, where OrderChain is higher by 0.9 and 1.2 ACC points respectively; this boundary case is consistent with the more subjective label structure of aesthetic rating. OLA also exceeds Ord2Seq in all 16 cells and NumCLIP in 15 of 16 cells while preserving the MLLM generation interface. These comparisons indicate that output-side ordinal alignment recovers much of the performance associated with backbone adaptation, without changing frozen MLLM parameters.

Table 1 shows the Designed Prompt sharpens last-token ordinal structure most on DR and HCI. To separate the lens contribution from the prompt contribution, Section 4.2 holds every trained OLA component fixed while varying only the inference-time prompt.

4.2 Component Ablation

Table 4 traces OLA across the design space identified by the diagnostics (full mean±std over five folds in Appendix D.4). Free supervised readouts on frozen hidden states (Probe Only, Capacity Probe) already recover much of the gap between Naive Prompt and OLA, confirming Section 2.2: the ordinal signal is available in hidden states and does not require a non-linear predictor. Probe Steering, however, stays far below the probe-based classifiers in most settings, even with a supervised probe-derived direction; this supports Section 2.3: improving a hidden-state direction alone does not suffice when the intervention still passes through the native output interface.

Comparing Capacity Probe with OLA-Offline tests whether the W_S-anchored lens and multi-layer aggregation preserve the useful signal while tying the readout to the digit-token interface. OLA-Offline is higher than Capacity Probe in 12 of 16 settings, with an average gain of 0.66 ACC points, useful but not the main source of the final improvement. The most consistent gain comes from reconnecting the fused lens distribution to the native logits: OLA-Online improves over OLA-Offline in all 16 settings (average +1.80 ACC), over Capacity Probe in all 16 (+2.45), and over Probe Only in all 16 (+3.48). These comparisons rule out the view that OLA is only a stronger offline classifier; the decisive step is the target-restricted correction that brings the lens distribution back into the MLLM generation process.

Single-Layer Logit further clarifies the roles of aggregation and online alignment. It outperforms simpler prompt or steering baselines but remains below OLA-Online because it uses only one validation-selected layer and lacks the discrepancy correction of Section 3.3. The full method is thus not just a probe, not just hidden-state steering, and not just single-layer logit substitution; it is an output-side alignment procedure that exploits probe-readable evidence while respecting the digit-token interface identified by the diagnostics.

The OL + Naive row tests prompt dependence directly: the full OLA pipeline trained under the Designed Prompt is applied at inference under the Naive Prompt without re-tuning the lens, fusion, or $\lambda^\star$. Across all 16 settings, OL + Naive substantially exceeds the Naive Prompt baseline (e.g., Qwen3-VL on DR: 0.794 vs. 0.688; LLaVA-NeXT on DR: 0.684 vs. 0.148) while remaining below

Backbone	Method	Adience		DR		HCI		Aesthetic		Mean ACC↑
		ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓	
Discriminative	Ord2Seq	0.639	0.430	0.842	0.250	0.609	0.520	0.644	0.304	0.684
	NumCLIP	0.651	0.410	0.831	0.270	0.696	0.350	0.665	0.296	0.711
Qwen2.5-VL 7B	Naive Prompt	0.572	0.544	0.156	1.567	0.278	0.952	0.372	0.772	0.345
	CAA	0.628	0.396	0.702	0.357	0.338	0.817	0.245	0.928	0.478
	OrderChain	0.765	0.282	0.870	0.140	0.653	0.347	**0.705**	**0.295**	0.748
	OLA	**0.809**	**0.233**	**0.920**	**0.080**	**0.685**	**0.315**	0.696	0.304	**0.778**
Qwen3-VL 8B	Naive Prompt	0.581	0.401	0.688	0.570	0.328	0.952	0.397	0.859	0.499
	CAA	0.648	0.374	0.722	0.333	0.358	0.792	0.265	0.904	0.498
	OrderChain	0.790	0.210	0.895	0.110	0.655	0.530	**0.710**	**0.294**	0.763
	OLA	**0.806**	**0.207**	**0.926**	**0.078**	**0.757**	**0.260**	0.698	0.302	**0.797**
Gemma-4 E4B	Naive Prompt	0.571	0.489	0.137	1.429	0.245	0.996	0.214	0.963	0.292
	CAA	0.574	0.437	0.744	0.307	0.319	0.896	0.341	0.816	0.495
	OrderChain	0.755	0.245	0.880	0.145	0.580	0.510	0.681	0.320	0.724
	OLA	**0.760**	**0.240**	**0.922**	**0.078**	**0.733**	**0.270**	**0.701**	**0.300**	**0.779**
LLaVA-NeXT 7B	Naive Prompt	0.450	0.766	0.148	1.891	0.216	1.043	0.307	0.984	0.280
	CAA	0.635	0.365	0.205	1.186	0.342	0.966	0.365	0.781	0.387
	OrderChain	0.682	0.367	0.839	0.221	0.622	0.438	0.696	0.324	0.710
	OLA	**0.791**	**0.229**	**0.924**	**0.076**	**0.723**	**0.308**	**0.719**	**0.281**	**0.789**

Table 3: Main benchmark results

OLA-Online. The gap to Naive Prompt lower-bounds the share of the OLA gain attributable to prompt-invariant latent ordinal evidence, and the gap to OLA-Online upper-bounds the contribution of the Designed Prompt at inference.

Finding 3. Target-restricted online alignment reliably converts latent evidence into ordinal decisions.

4.3 Training Cost and Parameter Budget

OLA reaches the accuracy reported in Table 3 at a small fraction of the cost of backbone adaptation. Only the lens and fusion parameters are trained, once per backbone-dataset pair, with the vision encoder, the language model, and the unembedding head including W_S held frozen. The gain over LoRA adaptation is therefore one of cost, not of avoided supervision. On the Adience dataset, OLA trains 37 to 110 times fewer parameters than the OrderChain LoRA adapters, and on Qwen3-VL it takes 0.575 GPU-hours against 19.80 on the same H200 GPU, while never backpropagating through the backbone. Most of this cost is a one-off frozen forward pass whose cached features are reused, so the trained component is a small readout on top of a frozen model. Appendix F gives the full comparison and the measurement protocol.

4.4 Robustness and Qualitative Analyses

Three additional analyses complement the main results: layer-window robustness of the lens, confusion-matrix structure under OLA, and layer-wise PCA of frozen hidden states.

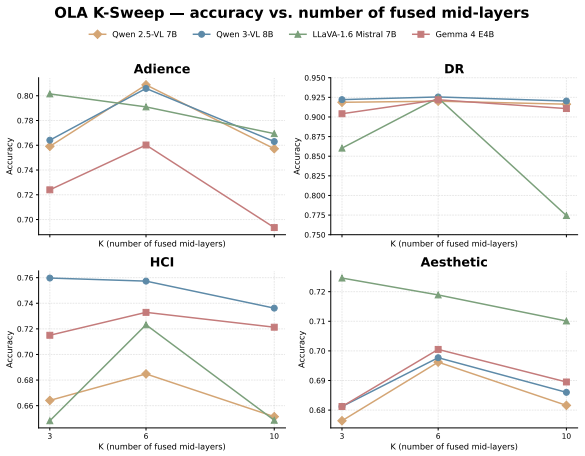


Figure 3: OLA accuracy under varying numbers of fused mid-to-deep decoder layers

Figure 3 reports OLA accuracy under varying numbers of fused mid-to-deep decoder layers. $K = 6$ is a stable default rather than a universal optimum: in most backbone-dataset curves, $K = 6$ is either best or close to best, while $K = 10$ becomes less stable when the window extends too broadly (most clearly LLaVA-NeXT on DR). Smaller windows are competitive in isolated cases (LLaVA-NeXT on Adience and Aesthetic) but lack the same overall consistency. We fix $K = 6$ throughout the main experiments without selecting K on the test set, supporting the role of multi-layer aggregation in Section 3.3: a stable readout window over the mid-decoder band rather than an oracle tuning

Backbone	Method	Adience		DR		HCI		Aesthetic		Mean ACC↑
		ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓	
Qwen2.5-VL 7B	Designed Prompt	0.601	0.399	0.830	0.227	0.483	0.610	0.454	0.681	0.592
	Probe Steering	0.650	0.364	0.725	0.331	0.386	0.749	0.429	0.666	0.548
	Probe Only	0.780	0.267	0.910	0.095	0.623	0.390	0.670	0.350	0.746
	Capacity Probe	0.792	0.272	0.894	0.082	0.616	0.401	0.687	0.329	0.747
	Single-Layer Logit	0.715	0.305	0.895	0.110	0.620	0.410	0.530	0.470	0.690
	OL + Naive	0.652	0.365	0.534	0.653	0.513	0.565	0.515	0.508	0.554
	OLA-Offline	0.792	0.259	0.872	0.096	0.625	0.382	0.687	0.323	0.744
	OLA-Online	**0.809**	**0.233**	**0.920**	**0.080**	**0.685**	**0.315**	**0.696**	**0.304**	**0.778**
Qwen3-VL 8B	Designed Prompt	0.627	0.401	0.840	0.176	0.274	0.987	0.430	0.812	0.543
	Probe Steering	0.670	0.343	0.745	0.307	0.406	0.724	0.449	0.643	0.568
	Probe Only	0.785	0.230	0.911	0.089	0.690	0.356	0.620	0.440	0.752
	Capacity Probe	0.781	0.220	0.913	0.082	0.716	0.293	0.646	0.374	0.764
	Single-Layer Logit	0.775	0.280	0.908	0.095	0.630	0.614	0.590	0.470	0.726
	OL + Naive	0.638	0.387	0.794	0.237	0.422	0.712	0.525	0.549	0.595
	OLA-Offline	0.800	0.210	0.916	0.080	0.726	0.282	0.653	0.369	0.774
	OLA-Online	**0.806**	**0.207**	**0.926**	**0.078**	**0.757**	**0.260**	**0.698**	**0.302**	**0.797**
Gemma-4 E4B	Designed Prompt	0.692	0.309	0.828	0.179	0.292	0.902	0.348	0.855	0.540
	Probe Steering	0.623	0.387	0.785	0.258	0.322	0.880	0.333	0.826	0.516
	Probe Only	0.735	0.275	0.905	0.108	0.701	0.360	0.680	0.330	0.755
	Capacity Probe	0.732	0.261	0.913	0.082	0.729	0.315	0.682	0.316	0.764
	Single-Layer Logit	0.725	0.290	0.895	0.120	0.650	0.410	0.620	0.430	0.723
	OL + Naive	0.684	0.320	0.493	0.705	0.392	0.772	0.563	0.512	0.533
	OLA-Offline	0.750	**0.231**	0.920	0.079	0.714	0.345	0.696	0.304	0.770
	OLA-Online	**0.760**	0.240	**0.922**	**0.078**	**0.733**	**0.270**	**0.701**	**0.300**	**0.779**
LLaVA-NeXT 7B	Designed Prompt	0.624	0.381	0.798	0.259	0.352	0.886	0.367	0.899	0.535
	Probe Steering	0.622	0.381	0.189	1.211	0.336	0.979	0.527	0.599	0.419
	Probe Only	0.755	0.302	0.840	0.175	0.702	0.350	0.705	0.301	0.751
	Capacity Probe	0.776	0.285	0.893	0.077	0.703	0.331	0.703	0.300	0.769
	Single-Layer Logit	0.690	0.315	0.821	0.257	0.604	0.456	0.530	0.550	0.661
	OL + Naive	0.593	0.427	0.684	0.460	0.368	0.857	0.492	0.659	0.534
	OLA-Offline	0.789	0.261	0.909	**0.071**	0.718	0.329	0.715	**0.275**	0.783
	OLA-Online	**0.791**	**0.229**	**0.924**	0.076	**0.723**	**0.308**	**0.719**	0.281	**0.789**

Table 4: Component ablation across the design space identified by the diagnostics.

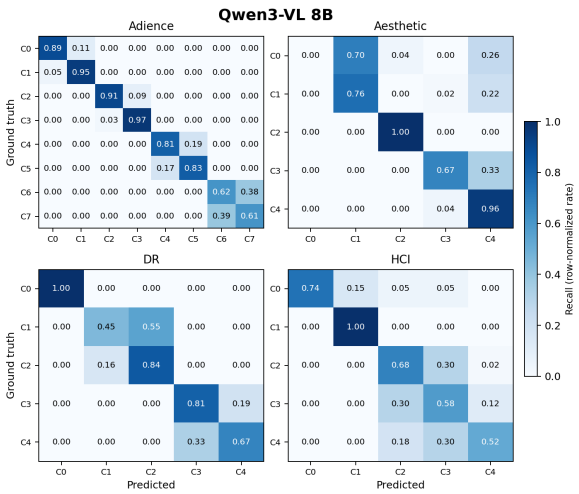


Figure 4: Row-Normalized Confusion Matrices

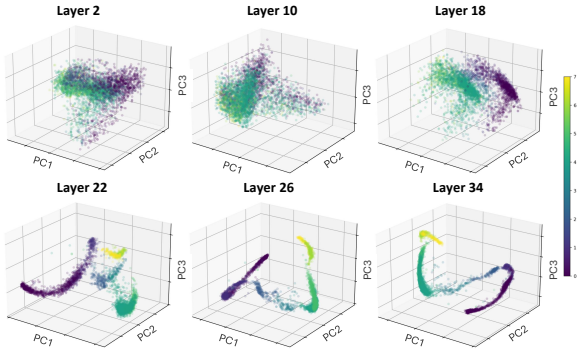


Figure 5: Layer-Wise PCA of Last-Token Hidden States

mechanism.

Figure 4 shows row-normalized confusion matrices of OLA on Qwen3-VL. Adience displays a strong near-diagonal pattern, with most residual errors confined to adjacent age groups. DR and HCI also concentrate errors among nearby ordinal classes, consistent with the MAE reductions in Table 3. This matters because ordinal prediction is not only a top-1 problem: confusing neighboring classes is less severe than confusing distant ones. Aesthetic remains less diagonal, matching its boundary-case role in the main benchmark and reflecting the noisier structure of subjective ratings. After OLA routes latent ordinal evidence into digit-token logits, residual errors tend to preserve ordinal proximity rather than becoming arbitrary label mistakes.

Figure 5 visualizes last-token hidden states from Qwen3-VL on Adience across decoder layers. Early layers show substantial class mixing and no clear monotonic color gradient. From the mid-decoder layers onward, the samples organize into a smoother class-ordered structure with a more visible color gradient. This pattern is consistent with the SRCC diagnostic in Table 1, where ordinal labels become more recoverable from intermediate and deeper hidden states. The visualization should be read qualitatively rather than as causal proof, since PCA only shows a low-dimensional projection; its role is to connect Finding 1 with the layer band used by OLA: the method reads from the part of the frozen MLLM where ordinal information is already organized, and the online correction then exposes it through the digit-token output interface. Cross-backbone PCAs on Adience and cross-dataset PCAs on Qwen3-VL appear in Appendix D.2.

5 Conclusion

Across four ordinal benchmarks and four MLLM backbones, latent ordinal evidence is linearly recoverable from frozen hidden states, yet the digit-token row space exposes less than 1.15% of this energy. OLA bridges the gap with W_S-anchored lenses and a target-restricted online correction, matching or exceeding LoRA-tuned baselines without modifying the frozen MLLM.

Limitations

We highlight three scope conditions of the present study.

First, our experiments cover four open-source MLLM backbones (Qwen2.5-VL, Qwen3-VL, Gemma-4, LLaVA-NeXT) and four ordinal benchmarks. Closed-source MLLMs (e.g., GPT-4V, Claude, Gemini) are not evaluated, since OLA requires access to mid-to-deep hidden states and the digit-token row submatrix W_S. The method applies directly to open-source models with standard transformer interfaces; extending to API-based models requires comparable logit access.

Second, all four benchmarks use single-digit ordinal label tokens ($C \leq 8$). Tasks with multi-digit ordinal outputs (e.g., direct year prediction, multi-digit severity scales) would require either label-tokenizer redesign or extending the discrepancy correction to token sequences. We view this as an orthogonal extension rather than a fundamental obstacle.

Third, OLA trains lightweight lens and fusion parameters per backbone-dataset pair (lens: $\mathcal{O}(Kdr)$ parameters; fusion: K scalars). Training operates on cached hidden states without backbone gradients, so the cost is much smaller than LoRA fine-tuning, but the method is not fully training-free. Lens transfer across datasets within the same backbone is left to future work.

References

Guillaume Alain and Yoshua Bengio. 2016. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*.

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, and 45 others. 2025a. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025b. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*.

Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. 2023. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*.

Axel Berg, Magnus Oskarsson, and Mark O'Connor. 2021. Deep ordinal regression with label diversity. In *2020 25th international conference on pattern recognition (ICPR)*, pages 2740–2747. IEEE.

Nicola Cancedda. 2024. Spectral filters, dark signals, and attention sinks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4792–4808.

Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. 2020. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325–331.

Jianlin Cheng, Zheng Wang, and Gianluca Pollastri. 2008. A neural network approach to ordinal regression. In *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*, pages 1279–1284. IEEE.

Raul Diaz and Amit Marathe. 2019. Soft labels for ordinal regression. In *Proceedings of the IEEE/CVF*

conference on computer vision and pattern recognition, pages 4738–4747.

Yao Du, Qiang Zhai, Weihang Dai, and Xiaomeng Li. 2024. Teach clip to develop a number sense for ordinal regression. In *European conference on computer vision*, pages 1–17. Springer.

Emma Dugas, Jared, Jorge, and Will Cukierski. 2015. Diabetic retinopathy detection. https://kaggle.com/competitions/diabetic-retinopathy-detection. Kaggle.

Eran Eidinger, Roee Enbar, and Tal Hassner. 2014. Age and gender estimation of unfiltered faces. *IEEE Transactions on information forensics and security*, 9(12):2170–2179.

John Hewitt and Christopher D Manning. 2019. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4129–4138.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *Preprint*, arXiv:2106.09685.

Zhipeng Jia, Xingyi Huang, I Eric, Chao Chang, and Yan Xu. 2017. Constrained deep weak supervision for histopathology image segmentation. *IEEE transactions on medical imaging*, 36(11):2376–2388.

Wanhua Li, Xiaoke Huang, Zheng Zhu, Yansong Tang, Xiu Li, Jie Zhou, and Jiwen Lu. 2022. Ordinalclip: Learning rank prompts for language-guided ordinal regression. *Advances in Neural Information Processing Systems*, 35:35313–35325.

Xiuyu Li, Jinkai Zhang, Mingyang Yi, Yu Li, Longqiang Wang, Yue Wang, and Ju Fan. 2026. Ets: Energy-guided test-time scaling for training-free rl alignment. In *Forty-third International Conference on Machine Learning*.

Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. 2024. Llava-next: Improved reasoning, ocr, and world knowledge.

Zhenxing Niu, Mo Zhou, Le Wang, Xinbo Gao, and Gang Hua. 2016. Ordinal regression with multiple output cnn for age estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4920–4928.

Frank Palermo, James Hays, and Alexei A Efros. 2012. Dating historical color images. In *European Conference on Computer Vision*, pages 499–512. Springer.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2023. Steering llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2023. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*.

Rossano Schifanella, Miriam Redi, and Luca Maria Aiello. 2015. An image is worth more than a thousand favorites: Surfacing the hidden beauty of flickr pictures. In *Proceedings of the international AAAI conference on web and social media*, volume 9, pages 397–406.

Xintong Shi, Wenzhi Cao, and Sebastian Raschka. 2023. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26(3):941–955.

Hossein Talebi and Peyman Milanfar. 2018. Nima: Neural image assessment. *IEEE transactions on image processing*, 27(8):3998–4011.

Gemma Team, Sherif El Abd, Vaibhav Aggarwal, Robin Algayres, Alek Andreev, Olivier Bachem, Ian Ballantyne, Cormac Brick, Victor Cărbune, Michelle Casbon, et al. 2026. Gemma 4 technical report. *arXiv preprint arXiv:2607.02770*.

Jinhong Wang, Yi Cheng, Jintai Chen, TingTing Chen, Danny Chen, and Jian Wu. 2023. Ord2seq: Regarding ordinal regression as label sequence prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5865–5875.

Jinhong Wang, Shuo Tong, Jian Liu, Dongqi Tang, Weiqiang Wang, Wentong Li, Hongxia Xu, Danny Z Chen, Jintai Chen, and Jian Wu. 2025. Orderchain: Towards general instruct-tuning for stimulating the ordinal understanding ability of mllm. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3477–3487.

Jiaqing Zhang, Sandeep Elluri, Bhanu Cherukuvada, Yonah Joffe, Jessica Sena, Miguel Contreras, Scott Siegel, Subhash Nerella, Catherine Price, and Parisa Rashidi. 2026. Auditing multimodal llm raters: Central tendency bias in clinical ordinal scoring. *arXiv preprint arXiv:2605.16386*.

A Datasets

This section documents the four ordinal benchmarks used in the main paper. For each dataset we give a brief overview that lists the source publication, the number of ordinal classes, and the per-class sample count, followed by the specific experimental settings.

A.1 Adience

Adience (Eidinger et al., 2014) is a benchmark for unfiltered face-image age estimation. The ordinal label space partitions human age into $C = 8$ ordered groups (0–2, 4–6, 8–13, 15–20, 25–32, 38–43, 48–53, 60+ years); boundaries are non-uniform in calendar years, which makes Adience a canonical benchmark for ordinal classification rather than ordinary multi-class age prediction. We follow the official five-fold protocol released with Adience. Per-class counts are reported in Table 5.

y	0	1	2	3	4	5	6	7
Range (yrs)	0–2	4–6	8–13	15–20	25–32	38–43	48–53	60+
Count	2,488	2,140	2,124	1,642	5,081	2,339	830	872

Table 5: Adience class distribution.

A.2 Diabetic Retinopathy (DR)

The Diabetic Retinopathy Detection corpus (Dugas et al., 2015) contains 35,126 retinal-fundus photographs graded on the five-level International Clinical Diabetic Retinopathy (ICDR) scale: 0 = no DR, 1 = mild non-proliferative DR (NPDR), 2 = moderate NPDR, 3 = severe NPDR, 4 = proliferative DR. We adopt a random five-fold split (75%/5%/20% for training, validation, and test), giving 28,101 training-plus-validation images and 7,025 test images per fold.

y	0	1	2	3	4
Grade	No DR	Mild	Mod.	Severe	Prolif.
Count	25,810	2,443	5,292	873	708

Table 6: DR class distribution.

A.3 Historical Color Images (HCI)

The Historical Color Image dataset (Palermo et al., 2012) is a 1,325-image benchmark for historical-photograph dating. Each photograph is annotated with the decade in which it was taken, giving $C = 5$ ordered classes (1930s, 1940s, 1950s, 1960s, 1970s). The corpus is exactly class-balanced (265 images per decade). We adopt a random five-fold split (75%/5%/20% for training, validation, and test).

A.4 Aesthetic

The Aesthetic dataset (Schifanella et al., 2015) is a 12,984-image collection covering four themes (Nature, Animal, Urban, People), annotated on a five-level scale that maps to the ordinal labels 0 = unacceptable, 1 = flawed, 2 = ordinary, 3 = professional, 4 = exceptional. Because the corpus is sharply imbalanced across classes, we adopt a class-balanced test partition.

y	0	1	2	3	4
Quality	Unaccept.	Flawed	Ordinary	Profess.	Except.
Count	23	552	7,603	4,574	232

Table 7: Aesthetic class distribution.

A.5 Cross-dataset summary

Table 8 consolidates the four datasets' ordinal class count, corpus size, per-fold test size, fold count, and split-protocol family.

Dataset	Attribute	C	Train+val	Test/fold	Folds
Adience	age	8	$\sim$ 14,900	$\sim$ 3,700	5 (official)
DR	disease grade	5	28,101	7,025	5 (random)
HCI	decade	5	1,060	265	5 (random)
Aesthetic	aesthetic rating	5	12,777	207	Balanced test

Table 8: Per-dataset split summary.

A.6 Probe-Site Protocol

To probe latent ordinal evidence, we extract hidden states at three diagnostic sites: (i) the mean of visual placeholder tokens with a neutral caption prompt, (ii) the assistant-start position under the same neutral prompt, and (iii) the assistant-start position under a task-designed prompt that specifies the ordinal scale and answer format. For each backbone, dataset, and diagnostic site, we fit per-layer linear probes on the training split and report the maximum test SRCC across layers as the diagnostic summary.

B Prompt Templates

This section presents the design principles of the Designed Prompts used to elicit ordinal answers from the four MLLM backbones. The four dataset-specific templates differ only in their domain content; they share the same four-component skeleton.

B.1 Design principles

Every Designed Prompt is composed of four components delivered in the following order: an AI role, a domain-knowledge background, an explicit task statement, and a coarse-to-fine reasoning step (CRD). The role primes the backbone with a domain-relevant persona; the background grounds the answer in the precise ordinal label scale; the task statement fixes the answer format; and the CRD step is the conditional chain-of-thought that the lens later reads from the hidden states.

Component (i), the AI role, is a one-sentence persona that primes the backbone with a domain-relevant identity. Component (ii), domain-knowledge background, is a short factual statement that enumerates the ordinal label scale of the dataset, including the semantic anchor of each class. Component (iii), the task statement, is a direct instruction that tells the backbone what to predict and in what format (a single digit, wrapped in a dataset-specific answer delimiter such as `$N$` or `&N&`). Component (iv), coarse-to-fine reasoning (CRD), is a two-step conditional chain that asks the backbone to first commit to a coarse super-class (a contiguous group of adjacent ordinal classes), and then commit to a fine class within that super-class. The coarse partition is a fixed, dataset-specific hierarchy that we define once and reuse across all four backbones; it is not derived from any test-time information, so the Designed Prompt contains no label leakage.

The four components together form a prompt that aligns the backbone's reasoning trace with the ordinal geometry that the lens reads at inference time. Empirically, replacing this prompt with the Naive Prompt (a bare class-mapping question without role, background, or coarse-to-fine reasoning) reduces last-token ordinal SRCC substantially in the deep layers, which is the diagnostic we report in the main paper.

B.2 Example template: Diabetic Retinopathy

The following Designed Prompt for the Diabetic Retinopathy (DR) dataset illustrates the four-component skeleton. The remaining three templates (Adience, HCI, Aesthetic) follow the same structure with domain-appropriate substitutions.

Role. You are an ophthalmologist trained to grade diabetic retinopathy from colour fundus photographs on the ICDR scale. **Background.** The ICDR scale defines five severity grades: 0 denotes no diabetic lesions; 1 mild NPDR with microaneurysms only; 2 moderate NPDR; 3 severe NPDR; 4 proliferative DR with neovascularisation. **Task.** Grade the DR severity in the fundus photograph and output a single digit from 0 to 4 wrapped as `&N&`. **Coarse-to-fine reasoning.** First commit to one of three coarse categories: Healthy for grade 0; Non-proliferative DR for grades 1 to 3; Proliferative DR for grade 4. State the visible lesion types that justify the coarse choice. If the coarse category is Non-proliferative DR, refine to mild 1, moderate 2, or severe 3 based on lesion count and distribution. Otherwise commit directly to 0 or 4. Output the final answer as `&N&`.

C Experimental Setup

This section consolidates the experimental settings for the two trained components that the appendix relies on: the per-layer SRCC probe used in the diagnostic of Appendix D, and the multi-layer ordinal lens used by OLA at inference.

C.1 SRCC probe

We extract the last-token hidden state from the frozen base backbone under the Designed Prompt with image input, L_2-normalise it, and fit a linear probe Linear$(d \to 1)$ per layer with mean-squared-error loss against the ordinal label. The optimiser is Adam with learning rate 10^{-4}, with cosine annealing over 100 epochs, batch size 128, and a fixed seed of 42. The reported metric is the Spearman rank-correlation between the probe's scalar prediction and the ordinal label. The probe head is selected by best test SRCC under a val-as-test protocol on Adience, DR, and Aesthetic; HCI uses its native validation split. Three backbones are probed: Qwen3-VL, Gemma-4, and LLaVA-NeXT.

C.2 Ordinal lens training

Training is in two stages, both with the MLLM backbone, the unembedding W_U, and its digit-token submatrix W_S strictly frozen; W_S integrity is verified by a SHA-256 hash before and after each stage. Inputs are the last-token hidden states cached from a single forward pass of the frozen MLLM at the $K{=}6$ mid-decoder layers $\mathcal{L}$. Per-layer channel-wise statistics (μ_ℓ, σ_ℓ) are computed once on the training split with σ_ℓ clamped to a floor of 10^{-3}, and the hidden state is standardised as $\tilde{h}_\ell = (h_\ell - \mu_\ell)/\sigma_\ell$ throughout.

Stage A: per-layer ordinal lens. For each layer $\ell \in \mathcal{L}$ we independently train the lens parameters $\{U_\ell, V_\ell, b_\ell\}$ that define $A_\ell = I + U_\ell V_\ell^\top$ with rank r=64. U_ℓ is initialised from $\mathcal{N}(0, 0.02^2)$ and V_ℓ is initialised to zero, so the initial transformation collapses to the identity. The objective is cross-entropy of $\sigma(W_S(A_\ell \tilde{h}_\ell + b_\ell))$ against the ordinal label. The optimiser is AdamW with learning rate 10^{-4}, batch size 128, gradient-clip norm 1.0, and a per-layer seed of $42 + \text{layer_id}$. Each layer trains for at most 100 epochs and the best-epoch parameters are kept under validation accuracy.

Stage B: softmax fusion. With all (A_ℓ, b_ℓ) frozen, we train the K fusion logits $\alpha \in \mathbb{R}^K$ such that $a_\ell = \sigma(\alpha)_\ell$. The objective is cross-entropy of $\sigma(\sum_\ell a_\ell W_S(A_\ell \tilde{h}_\ell + b_\ell))$ against the ordinal label. The optimiser is AdamW with learning rate 10^{-2}, no weight decay, batch size 128, and seed 42. Stage B trains for at most 30 epochs and the best-epoch α is kept under validation accuracy. After Stage B we cache $P^{\text{OLA}}(\cdot \mid x)$ on the train, validation, and test splits for downstream phases.

D Extended Experimental Results

This section reports a per-layer SRCC diagnostic across three MLLM backbones and the full main-benchmark and ablation tables with five-fold standard deviations.

D.1 Per-layer SRCC across backbones and datasets

We probe each layer of three MLLM backbones on all four datasets following the protocol in Appendix C.1, with Qwen3-VL having 36 decoder layers and hidden 4096, Gemma-4 having 42 layers and hidden 2560, and LLaVA-NeXT having 32 layers and hidden 4096.

SRCC peaks in the upper-middle decoder (L20–L28) and the peaks on Adience and Aesthetic fall inside the layer window $\mathcal{L} = \{16, 18, 20, 22, 24, 26\}$ used by OLA.

Layer	Adience	DR	HCI	Aesthetic
2	0.6845	0.7522	0.5771	0.7405
4	0.7283	0.7433	0.3927	0.7175
6	0.7154	0.7503	0.4278	0.7301
8	0.7423	0.7530	0.4769	0.7607
10	0.7259	0.7539	0.4980	0.7773
12	0.7632	0.7542	0.5234	0.8155
14	0.7966	0.7557	0.5467	0.8214
16	0.8589	0.7564	0.6977	0.8568
18	0.8989	0.7577	0.7627	0.9106
20	**0.9376**	**0.7578**	0.8148	0.9140
22	0.9353	0.7578	0.8377	0.9144
24	0.9344	0.7578	0.8235	**0.9163**
26	0.9336	0.7578	0.8342	0.9145
28	0.9337	0.7578	**0.8432**	0.9127
30	0.9342	0.7578	0.8271	0.9114
34	0.9347	0.7564	0.7983	0.9051

Table 9: Per-layer SRCC on Qwen3-VL.

Layer	Adience	DR	HCI	Aesthetic
2	0.7987	0.7485	0.3744	0.4907
9	0.8023	0.7492	0.2497	0.5670
12	0.8095	0.7541	0.2413	0.7309
15	0.8481	0.7554	0.5146	0.7909
18	0.8866	0.7564	0.7188	0.7934
20	0.8885	0.7563	0.7251	0.8610
21	0.8927	**0.7567**	0.7140	0.8662
22	0.9058	0.7565	0.7458	0.8874
24	0.9252	0.7560	0.7512	0.8960
26	0.9245	0.7562	**0.7728**	0.9083
28	0.9166	0.7561	0.7224	0.8748
33	0.9210	0.7559	0.6395	0.8864
36	0.9262	0.7561	0.6477	0.8976
40	**0.9293**	0.7563	0.6962	**0.9192**

Table 10: Per-layer SRCC on Gemma-4.

The Gemma-4 decoder peaks later than Qwen3-VL (L40 on Adience and Aesthetic, L26 on HCI), with DR plateauing at $\rho \approx 0.756$.

Layer	Adience	DR	HCI	Aesthetic
2	0.7751	0.7495	0.3611	0.6679
4	0.8164	0.7487	0.3857	0.7171
6	0.8304	0.7504	0.3736	0.6762
8	0.8504	0.7504	0.3742	0.6867
10	0.8608	0.7525	0.3568	0.6788
12	0.8666	0.7557	0.3250	0.6950
14	0.8950	**0.7559**	0.7214	0.8930
16	0.9164	0.7554	0.7960	**0.9056**
18	0.9149	0.7548	0.7921	0.8983
20	0.9136	0.7544	0.7784	0.8935
22	0.9152	0.7549	0.7891	0.9034
24	0.9163	0.7553	0.7974	0.9021
26	**0.9236**	0.7557	0.7960	0.8978
28	0.9216	0.7556	0.7961	0.8947
30	0.9192	0.7556	**0.8003**	0.9056

Table 11: Per-layer SRCC on LLaVA-NeXT.

LLaVA-NeXT carries a non-trivial ordinal sig-

nal already at L2 ($\rho \geq 0.66$ on three of four datasets) and peaks around L16–L30 across datasets. Table 12 summarises the best layer per backbone-dataset cell: the best-layer SRCC falls in $[0.756, 0.938]$ across all 12 cells and the best layer is consistently in the upper-middle of the decoder, never the final layer.

D.2 Layer-wise 3D PCA visualisations

We project the last-token hidden states onto their top three principal components at representative decoder layers and colour the points by ordinal class. Figures 6–12 show the resulting layer-wise structure: an ordinal arc emerges in the middle-to-late decoder layers, consistent with the per-layer SRCC peaks reported above.

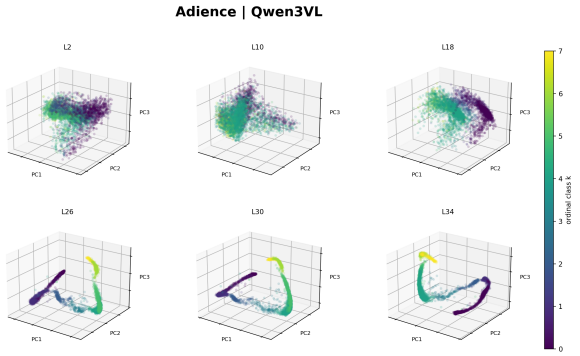


Figure 6: Layer-wise 3D PCA of last-token hidden states on Adience under Qwen3-VL. Points are coloured by the eight age groups; the ordinal arc emerges in the middle-to-late decoder layers.

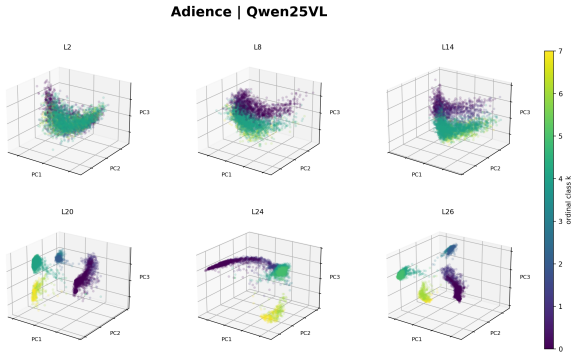


Figure 7: Layer-wise 3D PCA on Adience under Qwen2.5-VL.

D.3 Main results with five-fold standard deviations

Table 13 reports the main benchmark with mean ± std over five folds for Adience, DR, and HCI; Aesthetic uses one class-balanced fold.

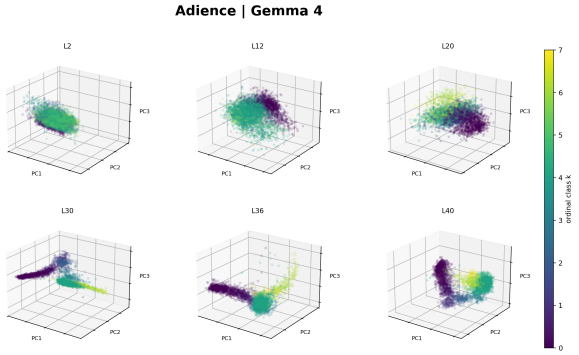


Figure 8: Layer-wise 3D PCA on Adience under Gemma-4.

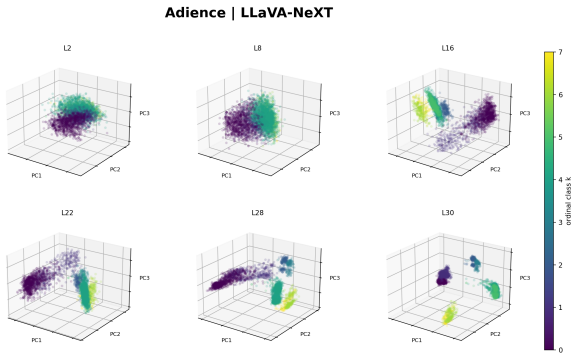


Figure 9: Layer-wise 3D PCA on Adience under LLaVA-NeXT.

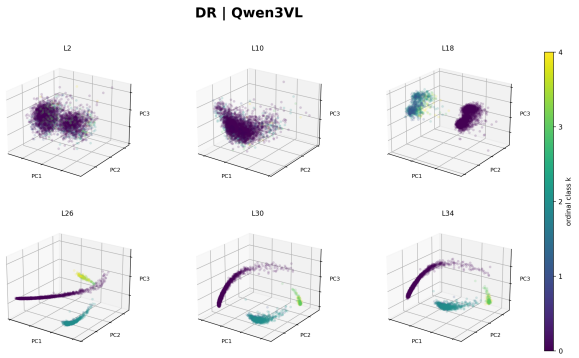


Figure 10: Layer-wise 3D PCA on DR under Qwen3-VL. Points are coloured by the five ICDR severity grades.

Backbone \ Dataset	Adience	DR	HCI	Aesthetic
Qwen3-VL	L20 / **0.9376**	L20 / 0.7578	L28 / 0.8432	L24 / 0.9163
Gemma-4	L40 / 0.9293	L21 / 0.7567	L26 / 0.7728	L40 / 0.9192
LLaVA-NeXT	L26 / 0.9236	L14 / 0.7559	L30 / 0.8003	L16 / 0.9056

Table 12: Best layer and best SRCC per backbone–dataset cell.

		Adience		DR		HCI		Aesthetic	
Backbone	Method	ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓
Qwen2.5-VL	Naive Prompt	0.5722 ±0.0168	0.5442 ±0.0644	0.1564 ±0.0058	1.5670 ±0.0118	0.2788 ±0.0027	0.9520 ±0.0044	0.3723	0.7721
	CAA	0.6284 ±0.1641	0.3956 ±0.3129	0.7022 ±0.0142	0.3574 ±0.0211	0.3381 ±0.0396	0.8167 ±0.0882	0.2453	0.9283
	OrderChain	0.7650 ±0.0344	0.2823 ±0.0127	0.8700 ±0.0098	0.1400 ±0.0016	0.6528 ±0.0423	0.3473 ±0.0223	**0.7053**	**0.2947**
	OLA (ours)	**0.8088** ±0.0156	**0.2327** ±0.0045	**0.9201** ±0.0040	**0.0799** ±0.0004	**0.6848** ±0.0281	**0.3153** ±0.0132	0.6962	0.3039
Qwen3-VL	Naive Prompt	0.5805 ±0.0340	0.4011 ±0.0299	0.6882 ±0.0121	0.5704 ±0.0229	0.3281 ±0.0201	0.9526 ±0.0202	0.3969	0.8588
	CAA	0.6484 ±0.1641	0.3743 ±0.2961	0.7222 ±0.0142	0.3334 ±0.0197	0.3582 ±0.0396	0.7921 ±0.0852	0.2653	0.9037
	OrderChain	0.7898 ±0.0252	0.2103 ±0.0067	0.8950 ±0.0043	0.1100 ±0.0005	0.6552 ±0.0315	0.5301 ±0.0251	**0.7101**	**0.2942**
	OLA (ours)	**0.8059** ±0.0206	**0.2068** ±0.0053	**0.9255** ±0.0022	**0.0780** ±0.0002	**0.7573** ±0.0409	**0.2602** ±0.0141	0.6977	0.3024
Gemma-4	Naive Prompt	0.5714 ±0.0179	0.4890 ±0.0314	0.1370 ±0.0046	1.4291 ±0.0227	0.2444 ±0.0161	0.9958 ±0.0256	0.2138	0.9627
	CAA	0.5743 ±0.0461	0.4373 ±0.1036	0.7441 ±0.0165	0.3071 ±0.0126	0.3191 ±0.0054	0.8957 ±0.0519	0.3412	0.8156
	OrderChain	0.7548 ±0.0312	0.2453 ±0.0093	0.8800 ±0.0086	0.1450 ±0.0014	0.5804 ±0.0258	0.5100 ±0.0227	0.6805	0.3196
	OLA (ours)	**0.7602** ±0.0649	**0.2399** ±0.0027	**0.9219** ±0.0059	**0.0782** ±0.0005	**0.7329** ±0.0250	**0.2700** ±0.0092	**0.7005**	**0.2995**
LLaVA-NeXT	Naive Prompt	0.4501 ±0.1295	0.7659 ±0.1685	0.1478 ±0.0101	1.8911 ±0.0277	0.2156 ±0.0221	1.0430 ±0.0251	0.3069	0.9842
	CAA	0.6350 ±0.0221	0.3651 ±0.0283	0.2050 ±0.0245	1.1861 ±0.1056	0.3422 ±0.0266	0.9663 ±0.1768	0.3654	0.7805
	OrderChain	0.6821 ±0.0506	0.3671 ±0.0272	0.8392 ±0.0161	0.2214 ±0.0042	0.6221 ±0.0371	0.4384 ±0.0262	0.6957	0.3243
	OLA (ours)	**0.7910** ±0.0410	**0.2293** ±0.0119	**0.9242** ±0.0083	**0.0759** ±0.0006	**0.7232** ±0.0553	**0.3083** ±0.0236	**0.7189**	**0.2812**

Table 13: Main results on four ordinal benchmarks under four MLLM backbones, with mean ± std over five folds (Adience, DR, HCI); Aesthetic uses one class-balanced fold.

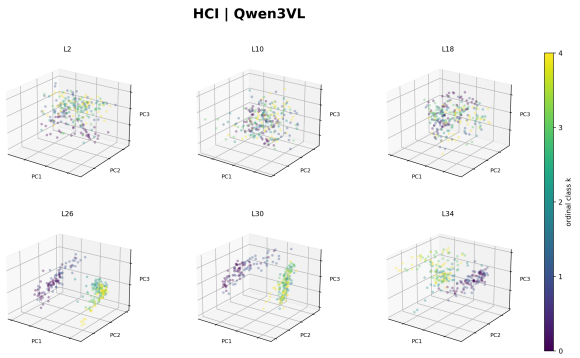


Figure 11: Layer-wise 3D PCA on HCI under Qwen3-VL. Points are coloured by the five decade classes.

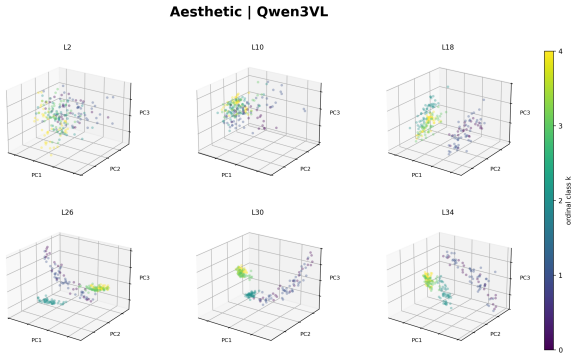


Figure 12: Layer-wise 3D PCA on Aesthetic under Qwen3-VL. Points are coloured by the five aesthetic grades.

D.4 Ablation studies with five-fold standard deviations

Table 14 reports the full ablation studies under four MLLM backbones on all four datasets. Eight methods are compared per backbone-dataset cell: Capacity Probe, Probe Only, Probe Steering, Single-Layer Logit, Designed Prompt, OL + Naive, OLA-Offline, and OLA-Online.

E Scaling Diagnostic at 32B

This section reports the diagnostic study of Section 2 on a 32B backbone, to test whether the interface-level failure mode identified by Findings 1 and 2 is specific to the 4B–8B scale of the main experiments.

E.1 Protocol

We use `Qwen3-VL-32B-Instruct` (dense, 64 decoder layers, hidden size $d = 5120$), a direct scale-up of the Qwen3-VL-8B backbone of the main paper, so that differences can be attributed to scale rather than to architecture. The probe protocol follows Appendix C.1, where last-token hidden states are taken under the Designed Prompt, a per-layer $\mathrm{Linear}(d \to 1)$ readout is trained with mean-squared error, and the peak test SRCC across decoder layers is reported. The retention R_{W_S} is computed as in Equation 3, using the probe direc-

Backbone	Method	Adience		DR		HCI		Aesthetic	
		ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓	ACC↑	MAE↓
Qwen2.5-VL	Capacity Probe	0.7923 ±0.0250	0.2721 ±0.0086	0.8941 ±0.0178	0.0821 ±0.0016	0.6161 ±0.0356	0.4009 ±0.0232	0.6871	0.3291
	Probe Only	0.7800 ±0.0286	0.2667 ±0.0098	0.9100 ±0.0106	0.0950 ±0.0011	0.6230 ±0.0251	0.3904 ±0.0157	0.6700	0.3500
	Probe Steering	0.6496 ±0.1851	0.3639 ±0.2881	0.7246 ±0.0071	0.3306 ±0.0196	0.3864 ±0.0619	0.7488 ±0.0806	0.4293	0.6658
	Single-Layer Logit	0.7150 ±0.0215	0.3050 ±0.0092	0.8950 ±0.0086	0.1100 ±0.0011	0.6200 ±0.0258	0.4100 ±0.0171	0.5300	0.4701
	Designed Prompt	0.6012 ±0.0280	0.3989 ±0.0180	0.8300 ±0.0054	0.2270 ±0.0015	0.4825 ±0.0248	0.6104 ±0.0314	0.4541	0.6812
	OL + Naive	0.6521 ±0.0626	0.3653 ±0.0250	0.5341 ±0.1525	0.6532 ±0.0760	0.5125 ±0.0531	0.5645 ±0.0314	0.5153	0.5080
	OLA-Offline	0.7918 ±0.0107	0.2593 ±0.0036	0.8722 ±0.0274	0.0963 ±0.0031	0.6252 ±0.0192	0.3821 ±0.0117	0.6867	0.3227
	OLA-Online	**0.8088** ±0.0156	**0.2327** ±0.0045	**0.9201** ±0.0040	**0.0799** ±0.0004	**0.6848** ±0.0281	**0.3153** ±0.0132	**0.6962**	**0.3039**
Qwen3-VL	Capacity Probe	0.7809 ±0.0316	0.2198 ±0.0089	0.9126 ±0.0102	0.0824 ±0.0009	0.7160 ±0.0562	0.2932 ±0.0231	0.6461	0.3740
	Probe Only	0.7851 ±0.0257	0.2300 ±0.0075	0.9107 ±0.0097	0.0894 ±0.0009	0.6900 ±0.0258	0.3564 ±0.0133	0.6200	0.4400
	Probe Steering	0.6696 ±0.1851	0.3431 ±0.2716	0.7446 ±0.0070	0.3065 ±0.0181	0.4064 ±0.0619	0.7244 ±0.0779	0.4493	0.6425
	Single-Layer Logit	0.7750 ±0.0172	0.2800 ±0.0062	0.9080 ±0.0034	0.0950 ±0.0004	0.6300 ±0.0258	0.4262 ±0.0252	0.5900	0.4700
	Designed Prompt	0.6274 ±0.0212	0.4009 ±0.0135	0.8398 ±0.0114	0.1755 ±0.0024	0.3742 ±0.0370	0.8511 ±0.0252	0.4300	0.8116
	OL + Naive	0.6381 ±0.0539	0.3872 ±0.0220	0.7941 ±0.0724	0.2366 ±0.0362	0.4215 ±0.0852	0.7115 ±0.0252	0.5248	0.5493
	OLA-Offline	0.7998 ±0.0378	0.2102 ±0.0099	0.9163 ±0.0120	0.0797 ±0.0011	0.7260 ±0.0413	0.2816 ±0.0161	0.6528	0.3692
	OLA-Online	**0.8059** ±0.0206	**0.2068** ±0.0053	**0.9255** ±0.0022	**0.0780** ±0.0002	**0.7573** ±0.0409	**0.2602** ±0.0141	**0.6977**	**0.3024**
Gemma-4	Capacity Probe	0.7324 ±0.0201	0.2605 ±0.0071	0.9130 ±0.0119	0.0822 ±0.0011	0.7294 ±0.0196	0.3152 ±0.0086	0.6817	0.3159
	Probe Only	0.7350 ±0.0215	0.2750 ±0.0080	0.9050 ±0.0060	0.1080 ±0.0007	0.7013 ±0.0200	0.3600 ±0.0103	0.6800	0.3300
	Probe Steering	0.6229 ±0.0677	0.3872 ±0.0916	0.7854 ±0.0087	0.2575 ±0.0106	0.3220 ±0.0028	0.8800 ±0.0509	0.3333	0.8261
	Single-Layer Logit	0.7250 ±0.0215	0.2900 ±0.0086	0.8950 ±0.0043	0.1200 ±0.0006	0.6500 ±0.0271	0.4100 ±0.0171	0.6200	0.4300
	Designed Prompt	0.6915 ±0.0460	0.3086 ±0.0195	0.8282 ±0.0150	0.1790 ±0.0032	0.2920 ±0.0275	0.9024 ±0.0850	0.3478	0.8551
	OL + Naive	0.6835 ±0.0638	0.3197 ±0.0240	0.4925 ±0.1242	0.7049 ±0.0621	0.3924 ±0.0742	0.7722 ±0.0500	0.5625	0.5119
	OLA-Offline	0.7497 ±0.0147	**0.2305** ±0.0046	0.9197 ±0.0064	0.0790 ±0.0006	0.7144 ±0.0274	0.3448 ±0.0132	0.6957	0.3043
	OLA-Online	**0.7602** ±0.0649	0.2399 ±0.0027	**0.9219** ±0.0059	**0.0782** ±0.0005	**0.7329** ±0.0250	**0.2700** ±0.0092	**0.7005**	**0.2995**
LLaVA-NeXT	Capacity Probe	0.7762 ±0.0586	0.2847 ±0.0216	0.8929 ±0.0345	0.0771 ±0.0031	0.7028 ±0.0132	0.3312 ±0.0062	0.7031	0.2995
	Probe Only	0.7550 ±0.0350	0.3021 ±0.0140	0.8400 ±0.0100	0.1750 ±0.0021	0.7023 ±0.0268	0.3500 ±0.0134	0.7053	0.3012
	Probe Steering	0.6217 ±0.0281	0.3808 ±0.0297	0.1893 ±0.0024	1.2113 ±0.1077	0.3355 ±0.0566	0.9788 ±0.1789	0.5266	0.5990
	Single-Layer Logit	0.6902 ±0.0248	0.3150 ±0.0113	0.8213 ±0.0208	0.2574 ±0.0065	0.6042 ±0.0302	0.4561 ±0.0253	0.5300	0.5500
	Designed Prompt	0.6244 ±0.0180	0.3813 ±0.0110	0.7982 ±0.0174	0.2590 ±0.0056	0.3523 ±0.0121	0.8861 ±0.0304	0.3671	0.8986
	OL + Naive	0.5934 ±0.0214	0.4269 ±0.0150	0.6835 ±0.1142	0.4598 ±0.0571	0.3676 ±0.0519	0.8569 ±0.0304	0.4924	0.6589
	OLA-Offline	0.7890 ±0.0524	0.2610 ±0.0173	0.9088 ±0.0138	**0.0712** ±0.0011	0.7180 ±0.0620	0.3288 ±0.0284	0.7146	**0.2754**
	OLA-Online	**0.7910** ±0.0410	**0.2293** ±0.0119	**0.9242** ±0.0083	0.0759 ±0.0006	**0.7232** ±0.0553	**0.3083** ±0.0236	**0.7189**	0.2812

Table 14: Ablation studies on four ordinal benchmarks under four MLLM backbones (mean ± std over five folds for Adience, DR, HCI; Aesthetic uses one class-balanced fold).

Dataset	peak SRCC↑		R_{W_S}		$\hat{R}_{W_S}$ (×chance)		native ACC↑	
	8B	32B	8B	32B	8B	32B	8B	32B
Adience ($C = 8$)	0.938	0.888	1.12%	0.38%	5.7	2.4	0.581	0.660
DR ($C = 5$)	0.758	0.811	1.06%	0.05%	8.7	0.5	0.688	0.723
HCI ($C = 5$)	0.843	0.853	1.08%	0.38%	8.8	3.9	0.328	0.348
Aesthetic ($C = 5$)	0.916	0.924	1.14%	0.06%	9.3	0.6	0.397	0.419

Table 15: Diagnostic results at 8B and 32B in the Qwen3-VL family. The 8B columns repeat the values of Table 1 (LT/D protocol), Table 2, and the Naive Prompt row of Table 3; the 32B columns are measured here on fold 0, and on fold 1 for Aesthetic. $\hat{R}_{W_S}$ is R_{W_S} divided by the $\mathrm{rank}(W_S)/d$ level a random subspace would reach, which is 0.195% and 0.122% at 8B and 0.156% and 0.098% at 32B. The 32B native prompt carries no coarse-label information and is therefore not word-for-word identical to the Naive Prompt of Section 4.1.

tion from the same run. The probe is trained on $N = 10{,}000$ examples, well above $d = 5120$.

E.2 Observations

Table 15 shows that both findings survive the scale-up. Ordinal evidence stays linearly recoverable, with peak SRCC between 0.811 and 0.924 and higher values at 32B than at 8B on DR and HCI. Retention in the digit-token row space falls on every dataset, and once the larger hidden size is accounted for it reaches chance level on DR and Aesthetic, so weak exposure is a structural property of the unembedding rather than a small-model artifact. Native digit-token accuracy improves only marginally and on all four datasets stays below what OLA already reaches on the 8B backbone in Table 3. Scaling the backbone is therefore not a substitute for output-side alignment. We do not train OLA at 32B and claim only that the bottleneck persists.

F Training Cost and Parameter Budget

This section reports the measurements behind Section 4.3. Table 16 compares OLA with the baselines of Table 3 on the Qwen3-VL-8B backbone, and Table 17 gives trainable parameter counts for the three backbones where both trained artifacts are available.

Method	Params	GPU-h	Bwd	Mem
Ord2Seq (Wang et al., 2023)	≈183–188 M	–	–	–
NumCLIP (Du et al., 2024)	≈86 M	–	–	–
OrderChain (Wang et al., 2025)	349.2 M	19.80	118,272	25.6
OLA (ours)	**3.17 M**	**0.575**	**0**	**18.4**

Table 16: Training cost on the Adience dataset with the Qwen3-VL-8B backbone. Params is the number of trainable parameters, GPU-h the single-GPU training time on one H200, Bwd the number of sample-level backward passes through the backbone, and Mem the peak memory in GB. OrderChain uses LoRA with $r = 128$ on all linear layers of the language model. The two discriminative references are trained outside the MLLM interface and we did not reproduce their timings on this hardware, so their compute columns are left empty.

		Trainable parameters		
Backbone	d	OLA	LoRA	ratio
Qwen3-VL 8B	4096	3,170,310	349,175,808	110.1×
Gemma-4 E4B	2560	1,981,446	72,613,888	36.6×
LLaVA-NeXT 7B	4096	3,170,310	335,544,320	105.8×

Table 17: Trainable parameters on the Adience dataset for the three backbones where both trained artifacts are available. The ratio column is the LoRA count divided by the OLA count. OrderChain uses LoRA with $r = 128$ and $\alpha = 256$. Total training FLOPs logged by the HuggingFace trainer are 5.20×10^{18}, 1.60×10^{18}, and 2.52×10^{18}.

G Pseudocode

Algorithms 1 and 2 give the pseudocode for the two OLA phases that match the method overview in Figure 2; lines that are pure I/O or device management are omitted for brevity.

H Related Work

H.1 Discriminative Ordinal Regression

Ordinal regression treats prediction as classification over ordered labels, where the distance between predicted and true classes is meaningful. Early work formulated this as a neural ranking problem (Cheng et al., 2008) and decomposed the C-class ordinal task into $C-1$ binary subproblems (Niu et al., 2016). Subsequent work introduced rank consistency through shared bias terms (CORAL, Cao et al., 2020) and conditional probabilities (CORN, Shi et al., 2023). Other approaches encode ordinal structure through soft labels that smooth class boundaries (Diaz and Marathe, 2019), label diversity that decorrelates probe directions (Berg et al., 2021), or weak supervision for medical grading (Jia et al., 2017). Domain-specific work includes face-age estimation (Eidinger et al., 2014) and neural image assessment (Talebi and Milanfar, 2018). Number-aware vision-language models such as NumCLIP (Du et al., 2024) and OrdinalCLIP (Li et al., 2022) introduce contrastive objectives that preserve numerical and ordinal relations in visual representations, and sequence-based formulations such as Ord2Seq (Wang et al., 2023) reduce ordinal classification to autoregressive sequence prediction. Our work extends this diagnostic perspective to frozen MLLM hidden states and the digit-token interface, rather than training a discriminative ordinal head.

H.2 Ordinal Prediction With MLLMs

Recent work applies MLLMs to ordinal recognition by reformulating ordered labels as digit-token generation. OrderChain is the closest baseline: it combines task-specific prompting with LoRA fine-tuning on age, aesthetic, and medical grading benchmarks (Wang et al., 2025). A separate line of work audits MLLMs as ordinal raters in clinical settings, reporting central-tendency bias on clinical ordinal scales (Zhang et al., 2026). Rather than updating the backbone, OLA asks whether the frozen model already contains ordinal evidence and whether the output interface exposes it, separating backbone adaptation from inference-time output alignment. The MLLM backbones we study span the Qwen-VL family (Bai et al., 2025a,b), the Gemma open-weights family (Team et al., 2026), and LLaVA-NeXT (Liu et al., 2024).

H.3 Linear Probing and Representation Geometry

Linear classifier probes (Alain and Bengio, 2016) measure how much task-relevant information a frozen representation linearly carries; structural probes (Hewitt and Manning, 2019) extend this to relational structure such as syntax. The linear representation hypothesis (Park et al., 2023) further argues that concepts in large language models are approximately encoded as linear directions in the residual stream. Our diagnostic uses the same linear-readout family and additionally measures the

Algorithm 1 Multi-layer Ordinal Lens Learning

Require: frozen MLLM M, dataset $\mathcal{D}$, designed prompt P, layer set $\mathcal{L}$, digit-row matrix W_S (frozen)
Ensure: lens parameters $\{A_\ell, b_\ell\}_{\ell\in\mathcal{L}}$, fusion logits α, cached $P^{\mathrm{OLA}}(\cdot \mid x)$

// one forward pass per sample, cache last-token hidden
register forward hooks on every layer $\ell \in \mathcal{L}$
for each $x \in \mathcal{D}$ **do**
  out $\leftarrow M.\textsc{forward}(P, \text{image}(x))$
  $H[x, \ell] \leftarrow$ captured last-token hidden at layer ℓ, for $\ell \in \mathcal{L}$
end for

// Stage A: per-layer ordinal lens (independent training)
for each $\ell \in \mathcal{L}$ **do**
  $(\mu_\ell, \sigma_\ell) \leftarrow$ mean and clamped std of $H[:, \ell]$ on train split
  initialise $A_\ell \leftarrow I + U_\ell V_\ell^\top$ with $U_\ell \sim \mathcal{N}(0, 0.02)$, $V_\ell \leftarrow 0$; $b_\ell \leftarrow 0$
  for epoch $= 1$ to 100 **do**
    for minibatch B **do**
      $\tilde{h} \leftarrow (H[B, \ell] - \mu_\ell)/\sigma_\ell$
      $\hat{z}_\ell \leftarrow W_S(A_\ell \tilde{h} + b_\ell)$
      $\mathrm{CE}(\mathrm{softmax}(\hat{z}_\ell), y[B]).\textsc{backward}()$; clip-norm 1.0; AdamW step
    end for
    track val accuracy; keep best-epoch (A_ℓ, b_ℓ)
  end for
end for

// Stage B: softmax fusion over K layers
freeze all $\{A_\ell, b_\ell\}$; $\alpha \leftarrow 0$; require_grad(α)
for epoch $= 1$ to 30 **do**
  for minibatch B **do**
    $\hat{z}_\ell(B) \leftarrow W_S(A_\ell \tilde{h}_\ell(B) + b_\ell)$ for $\ell \in \mathcal{L}$
    $a \leftarrow \mathrm{softmax}(\alpha)$
    $z^{\mathrm{OLA}} \leftarrow \sum_{\ell\in\mathcal{L}} a_\ell \, \hat{z}_\ell(B)$
    $\mathrm{CE}(\mathrm{softmax}(z^{\mathrm{OLA}}), y[B]).\textsc{backward}()$; AdamW step ($\alpha$ only)
  end for
  track val accuracy; keep best-epoch α
end for

// cache the instance-specific ordinal distribution
for split in {train, val, test} **do**
  $P^{\mathrm{OLA}}(\cdot \mid x) \leftarrow \mathrm{softmax}\big(\sum_\ell a_\ell \, \hat{z}_\ell(x)\big)$ for x in split
end for
return $\{A_\ell, b_\ell\}$, α, P^{OLA}

spectral and row-space exposure of these directions through the unembedding matrix.

H.4 Representation Steering and Parameter-Efficient Adaptation

Representation steering methods control model behavior by adding or projecting directions in the residual stream, with directions estimated from contrastive examples or learned probes (Panickssery et al., 2023). While these methods influence generation via hidden-state directions, they still rely on the native LM head, which our diagnostics show can be unreliable for ordinal prediction when a strong hidden direction is weakly exposed by the digit-token rows. OLA therefore applies alignment at the output side, modifying only candidate digit-token logits with a signal derived from W_S-anchored ordinal lenses. Parameter-efficient adaptation such as LoRA (Hu et al., 2021) provides an alternative route by training low-rank backbone updates; OLA differs by leaving the backbone frozen and training only output-side lens parameters.

H.5 Lens-Based Interpretation and Spectral Output Filtering

Logit-lens style methods map intermediate transformer states into vocabulary logits to analyze how predictions evolve across depth (Belrose et al., 2023). Spectral analyses of the unembedding matrix identify structured filters and dark signals that the output interface imposes on residual-stream content (Cancedda, 2024). OLA uses the same readout family but trains task-specific ordinal lenses anchored on W_S and uses their fused distribution for inference-time correction rather than interpretation only.

Algorithm 2 Inference-Time Ordinal Alignment

Require: frozen MLLM M, designed prompt P, trained lens $\{A_\ell, b_\ell\}$ and α, cached $P^{\mathrm{OLA}}(\cdot \mid x)$, base digit logits $L^{\mathrm{na}}[x]$ on val and test, grid Λ, digit-token ids $\mathcal{S}_{\mathrm{digit}}$, EOS ids $\mathcal{S}_{\mathrm{eos}}$
Ensure: $\lambda^\star$ and online-corrected predictions on test

// 1. select $\lambda^\star$ on validation (offline grid sweep)
for $\lambda \in \Lambda$ **do**
  for x in val **do**
    $Q(x) \leftarrow \mathrm{softmax}(L^{\mathrm{na}}[x])$
    $\omega(x) \leftarrow \max_c P^{\mathrm{OLA}}(c \mid x)$
    $\tilde{L} \leftarrow L^{\mathrm{na}}[x] - \lambda\,(Q(x) - P^{\mathrm{OLA}}(\cdot \mid x))\,\omega(x)$
    $\hat{y}(x) \leftarrow \arg\max_{c \in \mathcal{S}_{\mathrm{digit}}} \tilde{L}$
  end for
  record $\mathrm{ACC}_{\mathrm{val}}(\lambda)$
end for
$\lambda^\star \leftarrow \arg\max_\lambda \mathrm{ACC}_{\mathrm{val}}(\lambda)$

// 2. online generation on test (deterministic decoding)
register forward hooks on every $\ell \in \mathcal{L}$ that write into a shared dict H
for x in test **do**
  clear H ▷ per-sample reset
  prefill M with $(P, \mathrm{image}(x))$ ▷ hooks fill $H[\ell]$
  $P^{\mathrm{OLA}}(\cdot \mid x) \leftarrow \mathrm{softmax}\big(\sum_\ell a_\ell\, W_S(A_\ell H[\ell] + b_\ell)\big)$
  $\omega(x) \leftarrow \max_c P^{\mathrm{OLA}}(c \mid x)$
  greedily decode until reaching the answer slot ▷ no correction
  at the answer slot:
    $L_t \leftarrow M$.lm_head_logits
    $Q_t(c) \leftarrow \mathrm{softmax}(L_t[\mathcal{S}_{\mathrm{digit}}])$
    $L_t[\mathcal{S}_{\mathrm{digit}}] \leftarrow L_t[\mathcal{S}_{\mathrm{digit}}] - \lambda^\star\,(Q_t - P^{\mathrm{OLA}}(\cdot \mid x))\,\omega(x)$
    emit $\arg\max(L_t)$ as the ordinal prediction
  continue greedy decoding until EOS ▷ no correction
end for